

Project 3 Reflection

Responsive Design and Media Queries

This project helped me better understand the importance of responsive design and how it affects the overall user experience. Websites can be viewed on many different devices and screen sizes, so a layout that works well on a desktop may not work as well on a tablet or smartphone. Using the viewport meta tag allows the browser to display the page according to the device's actual screen dimensions. I also learned how media queries can change the layout at specific breakpoints. For this project, I used media queries to adjust the page for tablet and phone sizes. Testing the website using responsive design tools in Safari, Firefox, and Chrome helped me see how the layout changed at the different screen sizes and it helped me identify areas that had to be adjusted, despite not appearing problematic on the desktop view.

LESS Preprocessing Challenges

One of the biggest challenges I experienced was understanding how LESS preprocessing worked and how the .less source file related to the compiled .css file. I had to learn that the browser ultimately uses the compiled CSS rather than directly processing my LESS file. I used Node.js and the LESS compiler to compile project3.less into project3.css. Another challenge was getting the hamburger navigation to function correctly. Initially, the hamburger icon appeared, but clicking it did not display the navigation links. I was able to resolve this by creating a JavaScript file that adds and removes a CSS class when the menu button is clicked. Testing the navigation at different screen sizes helped me confirm it was working correctly.

Choosing LESS Features

I selected LESS features based on how they could make the stylesheet easier to organize and maintain. I used variables for colors, spacing, the maximum content width, and border radius so that these values could be changed in one location. I used a mixin for the styling shared by the call-to-action button. I also used a LESS function to lighten the primary color for the button's gradient. Operators were used for calculations involving the content width and spacing. These features allowed me to reduce repetitive CSS and demonstrate how preprocessing can make stylesheets more manageable.

Challenges for New Web Designers

Someone new to creating web pages, like myself, could encounter several challenges when completing a project like this. Understanding the relationship between HTML, CSS, LESS, JavaScript, and the compiled CSS file was initially confusing. In fact, response design was also challenging because the pages looked correct on the desktop but required significant adjustments on smaller screens. Learning how to use browser developer tools and responsive testing features helped identify these problems. I found that testing the project at the different screen sizes was one of the most important parts of making sure the final website worked as I intended it to do.

Page Validation

index.html

The screenshot shows a web browser window with the address bar at `validator.w3.org`. The page title is "Showing results for contents of text-input area (checked with vnu 26.9.9) - Nu Html Checker". The main heading is "Nu Html Checker". Below this, a note states: "This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change". The section "Showing results for contents of text-input area (checked with vnu 26.9.9)" contains a "Checker Input" box. This box has a "Show" menu with options: ☒ source, ☐ outline, ☐ image report, ☐ errors & warnings only, and an "Options..." button. The "Check by" dropdown is set to "text input" and "check as CSS" is unchecked. The input area contains the following HTML code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CMST 386 Projects</title>
  <!-- CSS Stylesheet -->
  <link rel="stylesheet" href="CSS/style.css">
</head>
<body>
  <header>
    <h1>CMST 386 Projects</h1>
  </header>
  <main>
```

Below the input area is a green "Check" button. At the bottom of the page, a green banner states: "Document checking completed. No errors or warnings to show." Below this banner, it says: "Used the HTML parser." and "Total execution time 2 milliseconds." At the very bottom, there are links: "About this checker" and "Report an issue".

project3.html

Showing results for contents of text-input area (checked with vnu 26.9....

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Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for contents of text-input area (checked with vnu 26.9.9)

Checker Input

Show

☐ source

☐ outline

☐ image report

☐ errors & warnings only

Options...

Check by

text input

☐ check as CSS

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CMST 386 Project 3 - Responsive Design</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="project3.css">
</head>
```

Check

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 5 milliseconds.

preprocessing.html

Showing results for contents of text-input ar...

Showing results for contents of text-input ar...

Showing results for contents of text-input a...

Nu Html Checker

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Showing results for contents of text-input area (checked with vnu 26.9.9)

Checker Input

Show

☐ source

☐ outline

☐ image report

☐ errors & warnings only

Options...

Check by

text input

☐ check as CSS

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CMST 386 Project 3 - LESS Preprocessing</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&display=swap" rel="stylesheet">

  <!--
  LESS preprocessing:
  I used the LESS compiler with Node.js to compile project3.less
-->
```

Check

Document checking completed. No errors or warnings to show.

Used the HTML parser.
Total execution time 5 milliseconds.

project3.css

 The W3C CSS Validation Service
W3C CSS Validator results for TextArea (CSS level 3 + SVG)

Jump to: Validated CSS

W3C CSS Validator results for TextArea (CSS level 3 + SVG)

Congratulations! No Error Found.

This document validates as [CSS level 3 + SVG](#) !

To show your readers that you've taken the care to create an interoperable Web page, you may display this icon on any page that validates. Here is the XHTML you could use to add this icon to your Web page:



```
<p>
  <a href="https://jigsaw.w3.org/css-validator/check/referer">
    
    </a>
  </p>
```